

1	The SAT scores of students are normally distributed with a mean of 500 and a standard deviation of 100. If a student scores 650, what is their z-score? a) -1.5 b) 0.5 c) 1.5 d) 2.0
2	A company finds that the time it takes to assemble a product is normally distributed with a mean of 45 minutes and a standard deviation of 5 minutes. Approximately what proportion of products take more than 55 minutes to assemble? a) 0.025 b) 0.05 c) 0.16 d) 0.34
3	Which of the following is true about a normal distribution? a) It has two modes. b) Its mean and variance are always equal. c) The probability density function is symmetric about the mean. d) Its standard deviation is always 1.
4	The curve of the normal distribution approaches the x-axis but never touches it. This property is known as: a) Continuity b) Symmetry c) Asymptotic nature d) Skewness
5	Which of the following best describes the z-score for a value in a normal distribution? a) Distance between the value and the mean divided by the variance b) Distance between the value and the mean divided by the standard deviation c) Standard deviation divided by the value d) Distance between the value and the mean
6	For the Gamma distribution, if the shape parameter $k=1$, the distribution becomes: a) Uniform distribution b) Normal distribution c) Exponential distribution d) Beta distribution
7	If $X \sim \text{Gamma}(3, 2)$, what is the mean of X? a) 3 b) 6 c) 9 d) 12
8	If $X \sim \text{Exponential}(\lambda)$, the cumulative distribution function (CDF) is given by: a) $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ b) $F(x) = e^{-\lambda x}$ for $x \geq 0$ c) $F(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ d) $F(x) = x e^{-\lambda x}$ for $x \geq 0$
9	A call center receives calls at an average rate of 3 calls per minute. What is the mean time between two successive calls? a) 3 minutes b) 1/3 minute c) 1 minute d) 1/9 minute

10	If the lifetime of a light bulb follows an exponential distribution with a mean of 1000 hours, the rate parameter λ is: a) 0.001 b) 1 c) 1000 d) 10
11	In a system where events occur according to an exponential distribution with rate $\lambda=4$, what is the probability of waiting less than 0.5 units of time for an event? a) $1-e^{-2}$ b) e^{-4} c) $1-e^{-4 \cdot 0.5}$ d) $e^{-4 \cdot 0.5}$
12	If $X \sim \text{Exponential}(3)$, what is the probability that X is between 1 and 2? a) $e^{-3}-e^{-6}$ b) $e^{-3 \cdot 1}-e^{-3 \cdot 2}$ c) $1-e^{-3 \cdot (1+2)}$ d) $1-(e^{-3 \cdot 1}+e^{-3 \cdot 2})$
13	The Poisson distribution is used to model: a) The probability of successes in a fixed number of trials. b) The probability of a given number of events occurring in a fixed interval of time or space. c) The waiting time between events. d) The sum of multiple random variables.
14	In a factory, the number of defective items produced per day follows a Poisson distribution with a mean of 5. What is the probability of producing more than 1 defective item in a day? a) $1-(P(X=0)+P(X=1))$ b) $P(X>1)=e^{-5}$ c) $P(X>1)=1-e^{-5}$ d) $P(X>1)=e^{-5}/2!$
15	The Poisson distribution is related to the exponential distribution because: a) The Poisson distribution models the waiting time between events. b) The exponential distribution is the sum of Poisson random variables. c) The Poisson distribution is the count of events in fixed intervals where the interarrival times follow an exponential distribution. d) Both distributions are discrete.
16	Which of the following is a condition for applying the Central Limit Theorem? a) The population must be normal. b) The sample size must be small. c) The sample size must be sufficiently large. d) The data must be discrete.
17	According to the Central Limit Theorem, the mean of the sampling distribution is equal to: a) The mean of the population. b) The variance of the population. c) Zero. d) One.
18	As per the Central Limit Theorem, for a sufficiently large sample size, the standard deviation of the sampling distribution is given by: a) σ^2 b) $\sigma/\sqrt{n}$ c) σ d) $\sigma \cdot \sqrt{n}$

19	The Central Limit Theorem applies to: a) Only populations with a normal distribution. b) Any population distribution, as long as the sample size is large enough. c) Only symmetric distributions. d) Only uniform distributions.
20	When can the sample size be considered "large enough" for the Central Limit Theorem to hold, for most distributions? a) $n > 10$ b) $n > 20$ c) $n > 30$ d) $n > 50$
21	Which of the following is true for the Central Limit Theorem? a) It only works for sample means. b) It applies to both sample means and sample sums. c) It applies only when the population variance is known. d) It requires the population to be finite.
22	The sampling error is defined as: a) The difference between the sample mean and the population mean. b) The difference between the sample variance and the population variance. c) The standard deviation of the population. d) The sum of all deviations from the sample mean.
23	The null hypothesis (H_0) generally states that: a) There is no significant difference or relationship. b) There is a significant difference or relationship. c) The sample data confirms the alternative hypothesis. d) The sample data does not follow any pattern.
24	Which of the following is true about the alternative hypothesis (H_1)? a) It states that there is no effect or difference. b) It is assumed to be true unless the evidence suggests otherwise. c) It is always tested first. d) It assumes that the null hypothesis is correct.
25	The significance level (α) in hypothesis testing is: a) The probability of rejecting the null hypothesis when it is true. b) The probability of accepting the null hypothesis when it is false. c) The critical value beyond which the null hypothesis is accepted. d) The margin of error in the sample mean.
26	In a two-tailed test, the rejection region for the null hypothesis is: a) Located on one side of the distribution. b) Split equally between both sides of the distribution. c) Located in the middle of the distribution. d) Non-existent.
27	A Type II error occurs when: a) The null hypothesis is rejected when it is true. b) The null hypothesis is accepted when it is false. c) The p-value is greater than α. d) The sample size is too large.
28	In hypothesis testing, the critical value is determined by: a) The sample mean. b) The p-value.

	c) The significance level and the type of test (one-tailed or two-tailed). d) The sample size only.
29	When conducting a Z-test, the null hypothesis is rejected if: a) The p-value is greater than the significance level. b) The sample mean equals the population mean. c) The Z-score is less than the critical value. d) The Z-score exceeds the critical value.
30	Which of the following is true about the Z-test? a) It is used when the population standard deviation is unknown. b) It is typically used for small sample sizes ($n < 30$). c) It assumes that the population is normally distributed. d) It is used for hypothesis testing of variances.
31	For a Z-test, if the computed Z-score is 2.5 and the critical value is 1.96, the decision is to: a) Accept the null hypothesis. b) Reject the null hypothesis. c) Fail to reject the null hypothesis. d) Increase the sample size.
32	The t-test is used when: a) The population standard deviation is known. b) The sample size is greater than 30. c) The sample size is small and the population standard deviation is unknown. d) The sample mean is known.

3 3	The degrees of freedom (df) for a one-sample t-test with a sample size of 25 is: a) 24. b) 23. c) 26. d) 25.
3 4	The F-test is used to compare: a) Two population means. b) Two population variances. c) The sample mean with the population mean. d) The difference between two sample means.
3 5	The F-distribution is: a) Symmetric and bell-shaped. b) Asymmetric and skewed to the right. c) Symmetric and follows a normal distribution. d) Symmetric and follows a chi-square distribution.
3 6	In an F-test, if the computed F-value is greater than the critical value, the decision is to: a) Accept the null hypothesis. b) Reject the null hypothesis. c) Fail to reject the null hypothesis. d) Increase the sample size.
3 7	The degrees of freedom for an F-test comparing two sample variances with sample sizes of 15 and 20 are: a) (14, 19). b) (15, 20). c) (14, 20). d) (20, 15).

3 8	The F-test is most commonly used in: a) Comparing two sample means. b) Comparing variances of two populations. c) Hypothesis testing for proportions. d) Testing the goodness-of-fit.
3 9	Which of the following assumptions is necessary for the F-test? a) The two populations should follow a normal distribution. b) The sample sizes must be equal. c) The two populations must be independent and identically distributed. d) The samples must have the same mean.
4 0	The degrees of freedom (df) for a two-sample t-test with sample sizes of 20 and 25 are: a) 44. b) 43. c) 45. d) 46.
4 1	In an F-test for comparing two variances, if the computed F-value is 2.3 and the critical F-value is 3.2, the decision is to: a) Accept the null hypothesis. b) Reject the null hypothesis. c) Fail to reject the null hypothesis. d) Perform a Z-test.
4 2	The Chi-Square test statistic is calculated by: a) $(\text{Observed frequency} - \text{Expected frequency}) / \text{Standard deviation}$. b) $(\text{Observed frequency} - \text{Expected frequency})^2 / \text{Expected frequency}$. c) $(\text{Expected frequency} - \text{Observed frequency}) / \text{Mean}$. d) $(\text{Observed frequency} + \text{Expected frequency})^2 / \text{Observed frequency}$.
4 3	In the Chi-Square goodness-of-fit test, the null hypothesis states that: a) There is no difference between observed and expected frequencies. b) The sample comes from a normal distribution. c) The sample means are equal. d) The variances are equal.
4 4	The Chi-Square distribution is: a) Symmetrical and bell-shaped. b) Positively skewed with only non-negative values. c) Negatively skewed with only non-positive values. d) Normally distributed.
4 5	The Chi-Square test is not appropriate when: a) The sample size is large. b) The expected frequencies are too small (less than 5). c) The observations are independent. d) The variables are categorical.
4 6	The Chi-Square distribution is most useful when the sample size is: a) Small. b) Large. c) Very large, especially when the expected frequencies are small. d) Any size.
4 7	Which of the following is NOT a requirement for using the Chi-Square test? a) Random sampling. b) Categorical data.

	c) A normal distribution. d) Sufficiently large sample size (expected frequencies ≥ 5).
4 8	In a Chi-Square goodness-of-fit test, the expected frequency for a category is 10, and the observed frequency is 6. The contribution to the Chi-Square statistic for this category is: a) 10. b) 16. c) 1.6. d) 1.0.
4 9	The Chi-Square test statistic is always: a) Negative. b) Zero. c) Positive. d) Normally distributed.
5 0	An estimator is said to be unbiased if: a) Its expected value is equal to the parameter it estimates. b) Its variance is minimized. c) Its mean squared error is minimized. d) It is consistent.
5 1	An estimator is said to be efficient if it: a) Has the smallest bias. b) Has the smallest variance among all unbiased estimators. c) Consistently approaches the true parameter value. d) Has the largest sampling variance.
5 2	An estimator is sufficient for a parameter if: a) It is unbiased. b) It contains all the information in the sample about the parameter. c) It has the smallest variance. d) It approaches the true value of the parameter as the sample size increases.
5 3	The Factorization Theorem is used to determine whether an estimator is: a) Consistent. b) Efficient. c) Unbiased. d) Sufficient.
5 4	Which of the following methods can be used to test sufficiency of an estimator? a) The Cramer-Rao bound. b) The likelihood ratio test. c) The factorization theorem. d) The MLE method.
5 5	An estimator is said to be consistent if: a) It is unbiased. b) It has minimum variance. c) It approaches the true value of the parameter as the sample size increases. d) It is efficient.
5 6	The Law of Large Numbers ensures that a sample estimator is: a) Unbiased. b) Efficient. c) Consistent. d) Sufficient.
5 7	If $\hat{\theta}$ is an estimator of the parameter θ and the expectation $E[\hat{\theta}] = \theta$, then the estimator is

	a) Biased b) Consistent c) Unbiased d) Efficient
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58	If the variance of an estimator $\hat{\theta}$ is 2 and the variance of an unbiased estimator of θ is 5, then: a) $\hat{\theta}$ is efficient. b) $\hat{\theta}$ is inefficient. c) $\hat{\theta}$ is unbiased. d) $\hat{\theta}$ is consistent. Answer: a) $\hat{\theta}$ is efficient.
59	Given that the likelihood function for a sample of size n is proportional to $\theta^n e^{-\theta x}$, the sufficient statistic for θ is: a) X b) X^2 c) $\sum_{i=1}^n X_i$ d) $\sum_{i=1}^n X_i^2$ Answer: c) $\sum_{i=1}^n X_i$
60	If $\hat{\theta} = \frac{X_1 + X_2}{2}$, where $X_1, X_2 \sim \mathcal{N}(\theta, 1)$, the bias of $\hat{\theta}$ is: a) 0 b) θ c) 1 d) 2θ Answer: a) 0
61	The likelihood function of a sample $X_1, X_2, \dots, X_n$ from a normal distribution is given by $L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(X_i - \mu)^2}{2\sigma^2}}$. The maximum likelihood estimator for μ is: a) $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$ b) $\hat{\mu} = \sum_{i=1}^n X_i$ c) $\hat{\mu} = X_1$ d) $\hat{\mu} = X_n$ Answer: a) $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$
62	If $X_1, X_2, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$, the unbiased estimator of σ^2 is: a) $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$ b) $\frac{1}{n-1} \sum_{i=1}^n (X_i - \mu)^2$ c) $\frac{1}{n} \sum_{i=1}^n X_i^2$ d) $\frac{1}{n-1} \sum_{i=1}^n X_i^2$ Answer: b) $\frac{1}{n-1} \sum_{i=1}^n (X_i - \mu)^2$

63	For a random sample $X_1, X_2, \dots, X_n$ from a Poisson distribution with parameter λ, the sufficient statistic for λ is: a) $\sum_{i=1}^n X_i$ b) $\sum_{i=1}^n X_i^2$ c) $\frac{1}{n} \sum_{i=1}^n X_i$ d) $\prod_{i=1}^n X_i$ Answer: a) $\sum_{i=1}^n X_i$
64	If $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$, the likelihood function is: a) $L(p) = p^n$ b) $L(p) = p^{\sum X_i} (1-p)^{n-\sum X_i}$ c) $L(p) = p^{\sum X_i}$ d) $L(p) = (1-p)^n$ Answer: b) $L(p) = p^{\sum X_i} (1-p)^{n-\sum X_i}$
65	For $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$, the maximum likelihood estimator of p is: a) $\frac{1}{n} \sum X_i$ b) $\frac{\sum X_i}{n}$ c) n d) $\sum X_i$ Answer: b) $\frac{\sum X_i}{n}$

66	For $X_1, X_2, \dots, X_n \sim U(0, \theta)$, the maximum likelihood estimator of θ is: a) $\min(X_i)$ b) $\max(X_i)$ c) $\frac{1}{n} \sum X_i$ d) X_1 Answer: b) $\max(X_i)$
67	For $X_1, X_2, \dots, X_n \sim \text{Poisson}(\lambda)$, the likelihood function is: a) $L(\lambda) = \lambda^{\sum X_i} e^{-n\lambda} / (\prod_{i=1}^n X_i!)$ b) $L(\lambda) = \lambda^n e^{-\lambda}$ c) $L(\lambda) = \lambda^{n\bar{X}} e^{-n\lambda}$ d) $L(\lambda) = \sum X_i e^{-\lambda}$ Answer: a) $L(\lambda) = \lambda^{\sum X_i} e^{-n\lambda} / (\prod_{i=1}^n X_i!)$
68	For $X_1, X_2, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$, the MLE of σ^2 is: a) $\frac{\sum (X_i - \mu)^2}{n}$ b) $\frac{\sum (X_i - \mu)^2}{n-1}$ c) $\frac{1}{n} \sum X_i$ d) $\frac{\sum X_i^2}{n}$ Answer: a) $\frac{\sum (X_i - \mu)^2}{n}$

69	For $X_1, X_2, \dots, X_n \sim \text{Exponential}(\lambda)$, the MLE of λ is: a) $\lambda = \frac{1}{\bar{X}}$ b) $\lambda = \frac{1}{n} \sum X_i$ c) $\lambda = \frac{n}{\sum X_i}$ d) $\lambda = X_1$ Answer: c) $\lambda = \frac{n}{\sum X_i}$